

Namma Pass

A peak-hour shared commute subscription that keeps every fare with the driver

Proposal for Namma Yatri (Moving Tech Innovations) | Case cities: Bengaluru, then Kochi | Prepared by Muhammed Sirfan C

The idea in one line Commuters on the same corridor and shift are grouped into threes (autos) or fours (cabs), share one regular driver in the peak hours, and split the fare. The driver collects about 1.4 times a solo fare, and Namma Yatri still takes no commission.

1. Why this belongs on Namma Yatri

Namma Yatri's public repository states its vision in five principles: zero commission, open, optimise, multimodal and shared transport. It also says growth should come without discounts and incentives. Most commuter products would break one of these. Namma Pass is designed to fit all of them.

Namma Yatri principle	How Namma Pass supports it
Zero commission, direct to driver	Riders split the fare and the driver collects all of it. Platform revenue stays the driver subscription.
Shared transport	Fewer vehicles carry the same commuters in the busiest hours, which means less congestion and lower cost per rider.
Sustainable growth, no incentives	No platform-funded discounts. Longer plans lock the seat price instead of cutting it.
Open	The design, the code and the pass metrics can be developed in the open and shown on the public data dashboard.
Multimodal	Pass groups can feed metro and bus stations, building on the metro last-mile work and the Namma Transit feature.
Driver first	A driver on a pass has the same riders every weekday, so the peak hours carry a predictable base income.

2. The problem

People who travel to the same place at the same time every day still book solo rides, wait for buses, or depend on informal carpools.

- **Cost:** a daily solo round trip is a large monthly expense.
- **Time:** buses and school vans make many stops, and the first passenger can spend an hour or more on board.
- **Reliability:** carpools depend on one private owner who can stop at any time.
- **Safety:** shared autos and vans are often overloaded. As a school student the author shared one auto with 12 other children.

3. The idea

- Riders enrol once with a home pin, a destination pin and pickup and return times.
- Each day is an ordinary Namma Yatri ride with planned stops. Pickup order and a fixed 3-minute wait per stop are set in advance.
- A cap on extra travel time (proposed: 15 minutes over a solo trip) stops the long first-passenger ride.
- The service runs only in the peak windows, about 7:00 to 10:00 and 15:00 to 18:00. Outside them, drivers take normal rides.

4. Economics under zero commission

Pricing rule: the group always pays 1.4 times the solo fare, split equally. A group of 4 pays 35% of the solo fare each, and a group of 3 pays about 47%. The driver collects all of it. Figures use a reference solo fare of ₹100 and are illustrative.

Per trip	Solo ride	Namma Pass cab (4 riders)	Namma Pass auto (3 riders)
Each rider pays	₹100	₹35	₹47
Driver collects	₹100	₹140	₹141
Commission to the platform	₹0	₹0	₹0
Vehicles for the commuters	3 or 4	1	1
Rider cost per month (44 trips)	₹4,400	₹1,540	₹2,068
Driver net per day, two peak trips, after daily subscription*	₹110 (cab) or ₹175 (auto)	₹190	₹257

*Daily subscription assumed at ₹90 for cabs and ₹25 for autos, as publicly reported. These would be replaced with the live values.

What Namma Yatri gains

- **More reliable driver activity.** A pass driver is on the road on every working day, so the daily subscription base is steadier.
- **Higher earnings for drivers** in the peak hours, with no incentive spend.
- **Retention.** Riders and drivers who share a daily routine are less likely to leave for another app.
- **A natural employer channel** (section 6) and a visible shared-transport product for the open-data dashboard.

Revenue options for the team to consider (none is assumed)

- Keep revenue exactly as it is: the driver subscription only.
- Offer an optional employer service plan, billed to the employer, for reports and group management. Riders and drivers pay nothing extra.

Payment design (open question)

Direct-to-driver is a core principle, so the default is that riders pay the driver directly by UPI on a fixed weekly or monthly schedule through the existing payment flow. A platform-held monthly wallet is simpler for riders but adds custody and compliance questions, so it is not recommended without the team's view.

5. Rules that keep it stable

- **Fixed schedule:** fixed pickup order and a fixed wait time per stop.
- **Leave and holidays:** no reduction. The seat stays reserved.
- **No mid-term refunds,** but days lost to a driver no-show, breakdown or platform error are credited to the next payment.
- **Fare lock:** 3 and 6-month plans lock the seat price for the term unless state fare guidelines change. This rewards commitment without a discount.
- **Replacement:** a rider can ask for a different driver or group, and every group has a named backup driver.

6. Employer plans and peak-window operations

Many employers already include a travel allowance in pay. Namma Pass lets a company cover all or part of its employees' seats.

- **Employer share:** the employer pays 100% (a travel benefit), a split such as 50/50, or nothing. The fare still goes to the drivers, settled directly in a monthly batch.
- **Driver stability:** employer-backed groups rarely lose members, so drivers keep the same trips and a predictable peak income.
- **Peak windows only:** 7:00 to 10:00 and 15:00 to 18:00. This keeps vehicles busy at the busiest times and leaves the rest of the day open for normal rides.
- **Women-only groups:** launch a group type with women drivers, building on Namma Yatri's programme to bring women into auto driving.

7. Pilot plan

Phase	Duration	What happens	Exit criteria
0. Demand test	3 weeks	In-app prompt to riders on four Bengaluru corridors: outer ring road, Manyata, Whitefield, Electronic City. No vehicles deployed.	Enough clustered demand for at least 10 groups on 2 corridors
1. Pilot	8 weeks	10 to 15 groups of autos and cabs on 2 corridors with regular and backup drivers.	Targets below met
2. Expand	3 months	All 4 corridors. Employer plans and women-only groups.	Stable renewals and driver earnings above solo
3. Next cities	After phase 2	Kochi (for example Infopark and the Kakkanad belt) and Chennai (OMR). School groups only after a safety and permit review.	Decision to scale city by city

Pilot KPI	Target
Groups at full strength	85% or more
On-time pickup	95% or more
Extra travel time versus a solo ride	15 minutes or less for every rider
Renewal after month 1	70% or more
Driver net earnings versus solo, peak hours	Higher, tracked weekly and published
Rider satisfaction	4.3 out of 5 or better

8. Risks

Risk	Mitigation
Rules on shared seats and school transport differ by state	Confirm permit requirements with the state transport department. Start with adult commuters.
A rider leaves and a group drops below its minimum	Corridor waitlist for replacements. Groups under 3 are not formed.
Fewer groups fill than expected	Demand survey first. Open only corridors with enough clustered demand.
Direct payments to drivers for monthly plans	Default to UPI directly to the driver on a fixed schedule. Escrow only if the team agrees.
Driver churn	Backup driver, replacement option and published driver earnings.

9. Building it on the open stack

Namma Yatri's code is public under the AGPL-3.0 licence. As described in its repository and documentation, it has a Haskell backend, PureScript mobile apps, a Rust location-tracking service, OSRM routing and Beckn/ONDC integration. Namma Pass would be built as additions to that stack:

Piece	Proposed approach
Recurring bookings	A pass entity that creates the daily rides, extending the existing Repeat Ride feature and the announced scheduled rides.
Group formation	A batch job that clusters pass applicants by home location and time window, using road travel times from the open routing engine.
Pickup order and tracking	Use the existing location-tracking service for pickup order, ETAs and the fixed wait timer.
Driver app	A roster view of the group with a fixed pickup order and attendance.
Open network	Expose the pass as a recurring fulfilment so other ONDC buyer apps can sell it. Payments stay direct to the driver.
Open data	Add pass metrics (groups, vehicles saved, driver earnings) to the public dashboard.

This mapping comes from the public repository structure and documentation. Details would be validated against the code before any design is final.

10. How the author would contribute

- **Weeks 1 to 2:** set up the backend locally, read the booking, scheduling and driver modules, and post a short design proposal in the project's GitHub Discussions.
- **Weeks 3 to 6:** build the grouping engine and demand-survey logic as a standalone service, share the results openly, and iterate with the team's feedback.
- **Then:** support the Bengaluru pilot, with metrics published on the open-data dashboard.

11. Working prototype

<https://muhammedsirfan.github.io/namma-pass/>

An interactive simulation of the grouping and pricing logic. Riders travel from five Bengaluru areas to a tech park on the outer ring road between 7:00 and 10:00. Controls: number of riders, extra-time cap, auto or cab, and employer share. All riders are synthetic and the geography is simplified.

Default run: 250 riders, 15-minute cap	Auto (3 riders)	Cab (4 riders)
Riders placed in groups	216 (34 waitlisted)	218 (32 waitlisted)
Groups formed	72, all full	65 (23 at 4 riders, 42 at 3)
Vehicles saved in the peak window	144	153
Average saving per trip	About 53%	About 58%
Rider bill per month versus solo	₹2,372 versus ₹5,085	₹2,114 versus ₹5,076
Driver net per peak day versus solo	₹298 versus ₹206	₹232 versus ₹140
Average worst extra travel time	About 9.5 minutes	About 11 minutes

Cab groups of four fill only when demand is dense, so the launch needs a demand survey first. Figures are illustrative and use an assumed fare formula.

12. About the author and next step

I am a 2026 B.Tech Computer Science graduate (Blockchain Technology) from VIT Vellore. I would like to build Namma Pass with your team, starting with a design proposal in the open, and would welcome a conversation about contributing as an engineer or intern.

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